

Capstone Project - MediAssist AI

Enterprise Healthcare Agentic AI Copilot

1. Problem Statement

Healthcare organizations store patient records, appointments, insurance data, and medical documents across multiple disconnected systems. This leads to fragmented information, slow retrieval, and inefficient workflows. MediAssist AI aims to provide a unified AI-powered healthcare assistant for intelligent information retrieval and decision support.

2. Stakeholders

- Doctors
- Hospital Administrators
- Front Desk Staff
- Insurance Team
- Clinical Operations Team
- Hospital Management

3. Main Pain Points

- Doctors: Scattered patient data, slow access to history and reports.
- Front Desk: Manual scheduling and patient lookup.
- Insurance Team: Slow claim verification and billing reconciliation.
- Administrators: Difficult reporting and compliance tracking.
- Clinical Operations: Workflow monitoring and escalation challenges
- Organization: Fragmented systems, knowledge silos, lack of unified AI interface.

4. Key Features Expected

- Conversational AI Assistant
- Enterprise RAG System
- Multi-Agent Architecture
- PostgreSQL Data Retrieval through MCP
- Multimodal AI Support
- Healthcare Analytics
- Workflow Automation
- LLM-as-a-Judge Evaluation
- Security Guardrails
- Azure Deployment with Docker

5. Data Sources

A. Unstructured Data:

Hospital SOPs, Clinical Workflows, Insurance Policies, Compliance Guidelines, Operational Manuals, Discharge Summaries, Patient Care Guidelines.

B. Multimodal Data:

Prescriptions, Medical Reports, Lab Report Images.

C. Structured Data:

Patients, Admissions, Appointments, Doctors, Departments, Billing, Lab Results, Prescriptions, Insurance Policies, Hospital Branches.

6. Sample User Queries

Operational:

- Show patient history for P10025
- Retrieve upcoming appointments
- Summarize latest lab reports

Analytical:

- Which department had the highest admissions this year?
- Compare appointment volume across hospital branches
- Analyze insurance claims by provider

Multimodal:

- Extract medicines from prescription image
- Summarize uploaded medical report
- Identify abnormalities in lab report image

Multi-Agent:

- Retrieve patient history, analyze lab results, summarize prescriptions, and generate recommendations.